Quick exposition of gauge invariance as a consequence of parameter invariance in Lagrangian mechanics and a peek at quantization.

"**Parameter invariance is gauge invariance for a one-dimensional world line.**"

My job in this notebook is to make that slogan clear, then to bridge it to a lot more physics.

Part I: From the steam age to quantum fields

Abstract

To understand nothing about the Standard Model of particle physics would be a shame, akin to never seeing Michelangelo's art or to never hearing Bach's music. The Standard Model is one of the greatest intellectual achievements of all time, and it also is one of the two most spectacularly verified theories in all of science, along with General Relativity. Nothing else even comes close to those two.

It is sobering to realize that we *really do know just how things work*: from quarks and gluons, electrons and photons, atoms and molecules, all the way to superclusters of galaxies. Even more sobering, the same kind of concise mathematics pertains to all of it, starting with the magical Lagrangian. Its depths reveal curvatures and invariants of abstract geometrical spaces that drive everything we know about the Universe. In this notebook, we hope to show that if you know just a bit of Classical Mechanics at a University level, you can get into the Standard Model to some depth. Both the Standard Model and General Relativity flow naturally from a magical reformulation of plain-old Newtonian mechanics, just $F = ma$.

Let's assume you are acquainted, even just superficially, with the Lagrangian and Hamiltonian approaches to Classical Mechanics, say from a course in Goldstein's book. If you've seen that a derivative of kinetic energy $\frac{1}{2} m v^2$ with respect to v gives $m v$, which equals momentum, that's probably enough to cross my bridge. I have posted several notebooks to the Wolfram Community with examples of the Lagrangian approach at work on the symmetrical spinning top and other scenarios. You should know a bit of multivariable calculus, too. I do *not* assume you're acquainted with tensors, differential geometry, or gauge theory. My job is to introduce you to these topics, showing how they follow directly from the Lagrangian approach.

There's no novel theory here, only newly concise and clear explanations, a hitch-hiker's guide to the staggeringly large and difficult literature of contemporary physics. I move you quickly from basic mechanics to the frontiers of quantum fields. All we need is the practical realization that *the labels we choose for points in any kind of space should not change the laws of physics*. I leave vast and steep abstractions to others, not to denigrate their value, but to cut to the chase.

By the end of this notebook, you should be able to

- **Demonstrate the Tensorial Nature of Motion**: show that the Euler-Lagrange equations are a ***covariant covector***, a purely geometrical object that fully captures the dynamics of a system across arbitrary definitions of coordinates and time.

- **Identify the Geometric Signature of Relativity**: explain why requiring a theory to be invariant with respect to choice of a time-evolution parameter forces a Lagrangian to be ***homogeneous of degree 1***.
- **Diagnose "Singular Systems"**: show that such homogeneity forces a ***singular Hessian matrix***, preventing us from solving for momenta in terms of velocities.
- **Invariance Implies Constants of Motion**: Derive both the ***relativistic mass-shell condition***, $p^2 = m^2$, and **the vanishing Hamiltonian** as direct consequences of parameter-invariance.
- **Anticipate Relativistic Quantum Fields**: be confident that we can extend these ideas from classical mechanics through to advanced physics, affording deeply rooted understanding from first principles.

Introduction

Actual physical motion should not depend on choice of coordinate system, within reason. Cartesian coordinates can be freely repositioned and rotated, yet still describe the same physics. Spherical and cylindrical coordinates routinely simplify problems. Classical textbooks like Morse and Feshbach describe many more useful coordinate systems, some highly esoteric and specialized.

- *Coordinate independence has its origin in cartography, with the effort to fit a spherical Earth onto a flat piece of mapping paper under this-or-that constraint: equal area, equal angles, straight lines for geodesics, and so on.*

Lagrangian mechanics takes coordinate independence to the max: we abstract *any* measurable quantities that constrain the ***kinematics*** of a system—the ways it *can* move—into ***generalized coordinates***. Angle of a wheel, length of a spring or a pulley cord, displacement of a joint on a sliding rod, and so on. Generalized coordinates, when chosen to honor constraints, eliminate accounting for reaction forces, making seemingly impossible problems tractable. Imagine describing the motion of the pistons, cranks, linkages, governors, and wheels of a steam locomotive purely via Newton's Laws in Cartesian coordinates. This same problem melts away and becomes even fun via the Lagrangian approach.

There really is no price to pay for the Lagrangian approach. The benefits compound as we progress from "homework problems" into relativity, field theory, quantum mechanics, and all the way to the Standard Model and General Relativity. But how?

The generalized coordinates for a specific problem actually describe a miniature Universe, a ***configuration manifold***, with its own idiosyncratic geometry and topology, its own curvatures and invariants. This fact unlocks the secret *geometric* power of the Lagrangian approach. Configuration manifolds succumb to the mathematics of ***differential geometry***.

Even from popular literature, we recognize *differential geometry* as the mathematics of cosmological gravitation via space-time curvature, and so it is. But the same framework also pertains to Lagrangian configuration manifolds. There we find curvatures and geodesics that constrain the ***dynamics*** of a system as it evolves and moves, analogous to the curvatures and geodesics that constrain gravitational motion in General Relativity.

Another slogan: **kinematics by choice of coordinates, dynamics by intrinsic geometry.**

- *“Intrinsic” here means precisely those facts of geometry that do not depend on choice of coordinates.*

Pressing on, we discover ***coordinate invariants***: integrals that do not depend on choice of coordinates. Think of an integral as a volume under a surface; its value will depend on units of measure, but not on whether we describe the surface in Cartesian or polar coordinates.

As a system evolves dynamically, its state, a sequence of coordinate values, traces out a ***world line***, a curved path through the configuration manifold. The coordinate invariants are properties, typically integrals, that remain constant of the world line regardless of coordinate system.

Naturally, we first think of Newtonian time for parametrizing dynamical progress along world lines. But suppose we *further* require that the Lagrangian not even depend on how we measure dynamical progress (within reason)—on what “clock” or ***dynamical parameter*** we choose. Relativity pops out of a ***parameter-invariant*** Lagrangian, almost without effort. Old Newtonian time becomes just another coordinate on the manifold, rather than a privileged, universal background parameter. The parameter that measures progress—the clock—becomes nearly arbitrary. In fact, it becomes a freely chosen ***gauge***.

So, by combining coordinate invariance with parameter invariance, we discover ***gauge invariance***, the gateway to advanced physics. **Parameter invariance is gauge invariance for a one-dimensional world line**. It is the first glimmer of further, wildly general gauge invariants that constitute the Standard Model.

- *The term “gauge” has its origin again in the steam age, where it refers to an arbitrary choice of the width of a rail line.*

In this notebook, we unlock the gateway and peek beyond. We’ll derive Einstein’s famous formula $p^2 = m^2$ (aka “$E = mc^2$,” though imprecisely) as an *algebraic* consequence of the unsolvability of Hamiltonian momenta in terms of coordinate velocities. We’ll end with a roadmap to the more general gauge theories of relativistic quantum fields by exactly the same procedures.

Contents

Index Notation

This book uses the old-fashioned index notation typified by Lovelock and Rund (L&R). It is written entirely in terms of transformation rules between coordinate systems. This is refreshingly concrete for the physicist concerned with calculation. After all, we must have numbers and arrays to do calculations. The downside of this ancient notation is that it's no help in understanding contemporary literature. We can't have our cake—coordinate-free notation for concise theorizing—and eat it—concrete coordinates for practical calculation—too. I'll occasionally make parenthetical remarks about this tension, but the meat of this notebook is in L&R's concrete notation.

We also avoid Cartan's exterior calculus, typified by the Grassmann wedge product, as in $dx \wedge dy$. While this is a beautiful and powerful formalism, it's not usually taught at an undergraduate level, and we don't want assume it of the reader. We also don't want to take the time and space to develop it. We can do all the physics we want without it, for now.

Ambiguity Warning

When ζ is a function of x and t, physicists and mathematicians sometimes write $\zeta(x, t)$. This notation is ambiguous because it also means "the value of the function ζ given specific coordinates x and t." We must tolerate this ambiguity in prose, and be very careful about it in code.

Points, Coordinates, States

Let X_n be an n-dimensional *differentiable manifold*. Informally, it is a space of abstract points p with Euclidean tangent spaces $T_p X_n$ at each point. Vectors live in the tangent spaces. There are also, at each point in X_n, corresponding ***dual co-tangent*** spaces, $T_p^* X_n$, where **covectors** live.

- *We elaborate and motivate the terms "vector" and "covector" below.*
- *$T_p X_n$, and $T_p^* X_n$ are indivisible notations, not some kind of T multiplying some kind of X.*
- *My definitions here are just touch points, introducing terms to aid the physicist's visual intuition. Find the mathematical rigor behind them in books on modern differential geometry, such as Chris Isham's book.*

Let the coordinates of points along some parametric curve $C(t)$ in X_n be values of similarly-named (eponymous) coordinate functions:

COORDINATE FUNCTIONS

$$x^j(\text{coordinate value}) = x^j(t)\ (\text{value of } \textit{coordinate function } x^j \text{ at } t),\quad j \in \{1, 2, \ldots, n\} \tag{1}$$

with "up" indices, and parameter t, initially Newtonian time. This dual meaning of x^i is another notational ambiguity we must tolerate in prose and be careful with in code.

A curve $C(t)$ of points in X_n represents a *mathematically possible* path—a continuous sequence of coordinate values $x^j(t)$—taken by a physical system as it evolves in time. A *physically possible* path is one such path, often the one with *least action*, but this notebook doesn't go deeply into this fact.

The manifold X_n represents the physical configuration of the system itself. The particular coordinate functions $x^j(t)$ depend on a choice of generalized coordinates in the Lagrangian language.

Derivatives along a Curve

Mathematically possible curves must be twice-differentiable with respect to t, with first derivatives

VELOCITY-VECTOR COMPONENTS

$$\overset{\bullet}{x}{}^j = \frac{d x^j}{dt} \equiv \overset{\bullet}{x}{}^j(t) \tag{2}$$

These are components of ***contravariant velocity vector*** (with "up" indices) along the curve $C(t)$. At each point $p = x(t)$ along the curve (x denoting the sequence $(x^1, x^2, \ldots, x^n)$) the velocity vector lives in the tangent space $T_{p=x(t)} X_n$ attached to the point p.

- *In some presentations, velocity vectors are directional-derivative operators in a **covariant operator basis** (with "down" indices): $v \equiv v^j \partial_j \equiv v^j \frac{\partial}{\partial x^j} \equiv \overset{\bullet}{x}{}^j \frac{\partial}{\partial x^j}$ via the **summation convention** (sum over repeated "up-down" pairs of indices in product expressions or in tensorial expressions). Mentally distinguish the vector, a geometric object in the tangent space, from its components in some basis. We are free to choose bases however we like, even something as exotic as a derivative operator.*
- *We elaborate on the terms "contravariant" and "covariant" below under the coordinate-transformation section.*
- *The summation convention is also called **contraction**.*
- *The components $\overset{\bullet}{x}{}^j(t)$ of velocities along the curve are the result of applying such directional derivative operators to the coordinate functions: $v^i \equiv \overset{\bullet}{x}{}^i = v^j(\partial x^i / \partial x^j) = v^j \delta^i_j$. In traditional notation, one would write $v = (v^1, v^2, \ldots, v^n) \bullet \nabla$, a dot product of numerical components with a gradient operator.*

- *These derivative operators should remind you of momentum operators in quantum mechanics.*
- *In coordinate-free presentations, tangent vectors are **equivalence classes of curves**. One abstractly defines addition of equivalence classes and multiplication of equivalence classes by scalars to make the equivalence classes resemble ordinary vectors as tuples of components. "Resemble," here, means "satisfy the axioms of a vector space." Thus, the equivalence classes become vectors in a rigorously mathematical sense.*

While the manifold X_n has n dimensions (n equals the number of degrees of freedom of the system), the curve $C(t)$ is a 1-dimensional object. This is why we can describe the kinematics and dynamics of a complex system like a steam engine using a single parameter t.

Lagrangian and Action

Let the Lagrangian $L\left(t, x^j, \dot{x}^j\right)$ be a twice-differentiable function of $2n+1$ independent variables, noting that x^j and $\dot{x}^j$ may be treated, at will, as variables or functions.

- *Treating the arguments of L as independent variables is justified because, as variables, they just describe kinematically (mathematically) possible states of the system. Dynamical evolution takes place along one specific curve* $C(t)$, the physically possible curve, *where the arguments are functions of* t.

Let the ***action integral*** along some curve C be

LOVELOCK ACTION INTEGRAL

$$I(C) = \int_{t=t_1}^{t=t_2} L\left(t, x^j(t), \dot{x}^j(t)\right) dt \qquad (3)$$

with $x^j(t)$ and $\dot{x}^j(t)$ here denoting the values of functions x^j and $\dot{x}^j$ at parameter values t along some curve $C(t)$.

- *The action is a scalar real number. It has different values for different curves.*
- *The* $2n+1$ *dimensionality will be critical to understanding gauge freedom. For now, the* $2n$ *generalized coordinates and velocities double as dynamical variables that double as functions of t, leading to equations of motion, and Newtonian time t is the privileged, universal parameter for tracking dynamical evolution of these variables. Later, we demote time t to be just another variable, opening the door to relativity, and replace its dynamical-tracking role with a freely chosen parameter* τ *as a gauge.*
- *This notebook is not concerned with solving the dynamical equations to find a "best" curve. Such is the subject of others of my notebooks. For now, just realize that the action integral is a function of the whole curve. Give me a curve C and I'll tell you the action* $I(C)$ *along that curve.*

Coordinate Transformations

To define *coordinate invariance* of the Lagrangian, we need results in this section.

Consider invertible, continuous, twice-differentiable coordinate-transformation functions to and from a new n-dimensional coordinate system $\overline{x}$, keeping the same parameter t (we'll change t later):

COORDINATE TRANSFORMATION FUNCTIONS

$$\bar{x}^j = \bar{x}^j(x^h) \qquad x^h = x^h(\bar{x}^j) \tag{4}$$

The derivatives $\dot{x}^h$ and $\dot{\bar{x}}^j$ transform ***contravariantly*** by the chain rule from the transformation functions:

COORDINATE TRANSFORMATION DERIVATIVES

$$\dot{x}^h = \frac{\partial x^h}{\partial \bar{x}^j}\dot{\bar{x}}^j \equiv \dot{x}^h\left(\bar{x}^j, \dot{\bar{x}}^j\right) \tag{5}$$

- ***Contravariance*** *of a tensorial (geometrical) object like the velocity vector with components* $\dot{x}^h$ *means that the magnitudes of components transform oppositely to the coordinate system. For example, Let* $\dot{\bar{x}}$ *(with the bar) measure a velocity in feet per second and* $\dot{x}$ *(without the bar) measure the same velocity in inches per second.* $\frac{\partial x}{\partial \bar{x}}$ *will be 12 inches per foot, but the magnitude of* $\dot{\bar{x}}$ *in feet per second will be 1 / 12 the magnitude of* $\dot{x}$ *in inches per second:* $\dot{x}(\text{ips}) = \frac{\partial x}{\partial \bar{x}}\left(12\,\frac{\text{inch}}{\text{foot}}\right)\dot{\bar{x}}(\text{fps})$, with the barred coordinates "on the bottom" multiplying the barred quantity.
- *A contravariant, rank-1 tensor is called, generically, a* ***vector****. Velocity is the canonical example. Vectors (at point p) inhabit the tangent space* $T_p X_n$.

From Equations 4 and 5, we get the rates of change of velocity-vector components $\dot{x}^h$ with respect to changes in the new coordinates $\bar{x}^j$:

$$\frac{\partial \dot{x}^h}{\partial \bar{x}^j} = \frac{\partial}{\partial \bar{x}^j}\frac{\partial x^h}{\partial \bar{x}^l}\dot{\bar{x}}^l = \frac{\partial^2 x^h}{\partial \bar{x}^j\,\partial \bar{x}^l}\dot{\bar{x}}^l \tag{6}$$

by renaming the dummy summation index from j to l: $\frac{\partial x^h}{\partial \bar{x}^j}\dot{\bar{x}}^j = \frac{\partial x^h}{\partial \bar{x}^l}\dot{\bar{x}}^l$.

We can swap the space and time derivatives to reveal the total time derivatives of the coordinate transformations as the same second-order partial derivatives:

TOTAL TIME DERIVATIVE OF THE COORDINATE TRANSFORMATIONS

$$\frac{\partial \dot{x}^h}{\partial \bar{x}^j} = \frac{\partial}{\partial \bar{x}^j}\frac{d x^h}{dt} = \frac{d}{dt}\frac{\partial x^h}{\partial \bar{x}^j} = \frac{\partial^2 x^h}{\partial \bar{x}^j\,\partial \bar{x}^l}\dot{\bar{x}}^l \tag{7}$$

- *TODO: justify this swapping via the chain rule, independence of x and* $\dot{x}$*, and Clairaut's theorem.*

We also get from Equation 5, by treating $\dot{\bar{x}}^j$ as a variable, and by the definition of partial derivative,

$$\frac{\partial \dot{x}^h}{\partial \dot{\bar{x}}^j} = \frac{\partial x^h}{\partial \bar{x}^j} \tag{8}$$

- *TODO: this feels dicey and glib. Make it rigorous.*

Coordinate Invariance of the Action

For the action integral, Equation 3, to be invariant with respect to coordinate transformations, it **suffices** to have

COORDINATE INVARIANCE

$$\overline{L}\left(t, \overline{x}^{j}(t), \overset{\bullet}{\overline{x}}^{j}(t)\right) = L\left(t, x^{h}\left(\overline{x}^{j}\right), \overset{\bullet}{x}^{h}\left(\overline{x}^{j}, \overset{\bullet}{\overline{x}}^{j}\right)\right) \tag{9}$$

with the j indices varying independently, e.g., $\overset{\bullet}{x}^{h}$ depends on all the $\overline{x}^{j}$ and on all the $\overset{\bullet}{\overline{x}}^{j}$, not just in matched pairs, and where $\overline{L}$ might be a different *form* from L, but the values of the two functions at the specified point (with the two different coordinates!) are equal.

- *TODO: prove* ***necessity****. Hint: invariance of the action integral must hold for any curve C and any endpoints t_1 and t_2. Also see Bolza's Lemma.*

This is shorthand for

$$\overline{L}\left(t, \overline{x}^{j}(t), \overset{\bullet}{\overline{x}}^{j}(t)\right) = L\left(t, x^{h}\left(\overline{x}^{j}(t)\right), \overset{\bullet}{x}^{h}\left(\overline{x}^{j}(t), \overset{\bullet}{\overline{x}}^{j}(t)\right)\right) \tag{10}$$

Differentiating with respect to $\overset{\bullet}{\overline{x}}^{j}$, treating it as an independent variable, we get

COVARIANT CONJUGATE MOMENTA

$$\frac{\partial \overline{L}}{\partial \overset{\bullet}{\overline{x}}^{j}} = \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial \overset{\bullet}{x}^{h}}{\partial \overset{\bullet}{\overline{x}}^{j}} = \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial x^{h}}{\partial \overline{x}^{j}} \tag{11}$$

The $\frac{\partial L}{\partial \overset{\bullet}{x}^{h}}$ are manifestly covariant, with the “up” index in the “denominator” of the derivative’s becoming a “down” index overall. Also carefully note $\overline{L}$ on the left and L on the right.

- *Of course,* $\frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \equiv p_h$ *are the* ***canonical*** *or* ***conjugate momenta*** *of Hamiltonian theory*.

We also get

PARTIAL DERIVATIVE OF L wrt COORDINATE

$$\frac{\partial \overline{L}}{\partial \overline{x}^{j}} = \frac{\partial L}{\partial x^{h}} \frac{\partial x^{h}}{\partial \overline{x}^{j}} + \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial \overset{\bullet}{x}^{h}}{\partial \overline{x}^{j}} = \frac{\partial L}{\partial x^{h}} \frac{\partial x^{h}}{\partial \overline{x}^{j}} + \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial^{2} x^{h}}{\partial \overline{x}^{j}\, \partial \overline{x}^{l}} \overset{\bullet}{\overline{x}}^{l} \tag{12}$$

- *TODO: this expression looks suspiciously like a* ***covariant derivative*** *with the highlighted term a* ***connection*** *or Christoffel symbol. Is this a deep fact? (hint: yes)*

The components $\frac{\partial \overline{L}}{\partial \overline{x}^{j}}$ are not tensorial. Let’s construct a proper *covariant* tensor. We’ll look to eliminate the highlighted term, which includes a second partial derivative of the coordinate transformation, by finding it in another expression.

Start with the total time derivative of another *covariant* quantity, the conjugate momenta, in the transformed coordinates, by the product rule and by substituting the total time derivative of the transformation functions from Equation 7:

$$\frac{d}{dt}\left(\frac{\partial \overline{L}}{\partial \overset{\bullet}{\overline{x}}^{j}}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial x^{h}}{\partial \overline{x}^{j}}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}^{h}}\right) \frac{\partial x^{h}}{\partial \overline{x}^{j}} + \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{d}{dt}\left(\frac{\partial x^{h}}{\partial \overline{x}^{j}}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}^{h}}\right) \frac{\partial x^{h}}{\partial \overline{x}^{j}} + \frac{\partial L}{\partial \overset{\bullet}{x}^{h}} \frac{\partial^{2} x^{h}}{\partial \overline{x}^{j}\, \partial \overline{x}^{l}} \overset{\bullet}{\overline{x}}^{l} \tag{13}$$

We can now subtract out the offending second-order partial derivative by subtracting the $\frac{\partial \overline{L}}{\partial \overline{x}^{j}}$ of Equation 12:

EULER-LAGRANGE TRANSFORMATION LAW

$$\frac{d}{dt}\left(\frac{\partial \overline{L}}{\partial \overset{\bullet}{\overline{x}}{}^{j}}\right)-\frac{\partial \overline{L}}{\partial \overline{x}^{j}}=\frac{\partial x^{h}}{\partial \overline{x}^{j}}\left(\frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}{}^{h}}\right)-\frac{\partial L}{\partial x^{h}}\right) \tag{14}$$

The highlighted quantities transform covariantly, i.e., by multiplying barred quantities by the Jacobian of the inverse coordinate transformation functions, as can be seen even more clearly in the following inversion:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}{}^{h}}\right)-\frac{\partial L}{\partial x^{h}}=\frac{\partial \overline{x}^{j}}{\partial x^{h}}\left(\frac{d}{dt}\left(\frac{\partial \overline{L}}{\partial \overset{\bullet}{\overline{x}}{}^{j}}\right)-\frac{\partial \overline{L}}{\partial \overline{x}^{j}}\right) \tag{15}$$

- *The result is beautiful, but this derivation takes a bit of squinting, and it's difficult to imagine how someone came up with it. Does coordinate-free presentation make it more concise or obvious? (hint: more concise, yes; more obvious, no).*
- ***Covariance** of a tensorial object (with "down" indices) means that the magnitudes of components transform along with the coordinate system. For example, Let $\overline{T}$ measure a temperature gradient in degrees per foot and T measure the same gradient in degrees per inch. $\frac{\partial x}{\partial \overline{x}}$ will be 12 inches per foot. The magnitude of $\overline{T}$ in degrees per foot will be 12 times the magnitude of T in degrees per inch, i.e.,* $\overline{T}(\text{dpf}) = \frac{\partial x}{\partial \overline{x}}\left(12\,\frac{\text{inch}}{\text{foot}}\right) T(\text{dpi})$, with unbarred coordinates on top, multiplying the unbarred T.
- *A covariant, rank-1 tensor is called, generically, a **covector**. A gradient is the canonical example. Covectors (at point p) inhabit the cotangent space $T^*{}_p\, X_n$.*
- *Recall how we construct a directional derivative by **contracting**, or summing over repeated up-down indices, by contracting the components v^j of a contravariant vector v with the covariant basis of partial-derivative operators $\frac{\partial}{\partial x^j}$, which is a gradient operator.*

Euler-Lagrange Covector

The highlighted quantities in Equation 14 should "ring a bell" as the terms of the Euler-Lagrange equations of motion. You will usually find these as the result of a lengthy derivation in the *calculus of variations*. But here, we found them simply by searching for a tensorial version of a derivative of L! We didn't need to introduce the calculus of variations, beautiful as it is.

These quantities deserve a name: The ***Euler-Lagrange covector*** (the "up" indices in the denominator (lower) position of derivatives become "down" indices in the result, and "down" indices are the sigil of a covector):

EULER-LAGRANGE COVECTOR

$$E_h(L) \equiv \frac{d}{dt}\left(\frac{\partial L}{\partial \overset{\bullet}{x}{}^{h}}\right)-\frac{\partial L}{\partial x^{h}} \tag{16}$$

Summary of Coordinate Invariance

Quantity	Notation	Tensorial Rank	Transformation Type
Coordinate	x^h	(1, 0)	Contravariant
Velocity	$\dot{x}^h$	(1, 0)	Contravariant
Momentum	$p_h = \frac{\partial L}{\partial \dot{x}^h}$	(0, 1)	Covariant
E - L Covector	$E_h (L)$	(0, 1)	Covariant

Physical Interpretation

$E_h(L)$ **vanishes for physically realistic motion.** With that vanishing, we treat the coordinates and velocities as functions of the parameter, and thus get differential equations for the motion. In this notebook, we do not solve dynamical problems, and thus don't assume that these quantities vanish. We have only shown that **the $E_h(L)$ transform covariantly when the Lagrangian is invariant with respect to coordinate transformations**. However, the classical laws of motion described by $E_h(L) = 0$ are also independent of the coordinate system, given that the coordinates are adequate to distinguish points along the curve C. **The laws of motion are also a purely geometric property of the curve itself.** The laws of motion follow from finding geodesics (shortest paths) in a curved space with $E_h(L)$ measuring the deviation of the curve from a geodesic.

Parameter Invariance of the Action

Now that we know that neither the kinematics nor the dynamics depend on how we define coordinates on the configuration manifold, let's make the physics not even depend on how we measure dynamical progress—on the choice of clock. In the offing, we demote our old Newtonian time to be just another coordinate. Relativity will be inevitable.

Consider invertible parameter transformations, independent of the coordinate transformations of Equation 4:

$$\tau = \tau(t);\ \ t = t(\tau) \tag{17}$$

The action integral (Equation 3) is parameter-invariant if and only if

PARAMETER INVARIANT ACTION INTEGRAL

$$\int_{t_1}^{t_2} L\left(t, x^j(t), \dot{x}^j(t)\right) dt = \int_{\tau_1}^{\tau_2} L\left(t(\tau), x^j(t(\tau)), \frac{d x^j(t(\tau))}{d\tau}\,\frac{d\tau}{dt}\right)\frac{dt}{d\tau}\, d\tau \equiv \int_{\tau_1}^{\tau_2} L\left(t(\tau), x^j(t(\tau)), \grave{x}^j\,\frac{d\tau}{dt}\right)\frac{dt}{d\tau}\, d\tau \tag{18}$$

where $\tau_1 = \tau(t_1)$ and $\tau_2 = \tau(t_2)$ and introducing the over-tick notation $\grave{x}^j \equiv \frac{d x^j(t(\tau))}{d\tau}$. This is an exact identity by straight substitution, by the chain rule to extract $d\tau / dt$ in the last argument slot of L, and by a scaling of the differentials $dt = \frac{dt}{d\tau}\, d\tau$. It does not entail a new function $\overline{L}$ as we might need with coordinate transformations.

- *Insist that $\frac{dt}{d\tau} > 0$, so that the inverse derivative $\frac{d\tau}{dt}$ exists and so that both parameters go "forward" under the transformation. This condition lets both t and τ model physical time. The transformation can only stretch or compress time, not flip it or stall it.*
- *We usually choose τ to be **relativistic proper time**, defined below.*

The next objective is to convert the parameter-invariant action integral into an identity that has similar functional forms on both sides, but also maintains numerical equality, again necessary and sufficient for the two integrals to be equal:

$$\int_{t_1}^{t_2} L\left(t, x^j(t), \dot{x}^j(t)\right) dt = \int_{\tau_1}^{\tau_2} L\left(\tau, x^j(\tau), \dot{x}^j\right) d\tau \tag{19}$$

Notice again that L is the same on both sides of this equation. No new $\overline{L}$ function here.

Turns out this is true if and only if L is ***homogeneous of degree 1*** in the velocity argument, meaning that

HOMOGENEITY OF L

$$L\left(\tau, x^j(\tau), \dot{x}^j \frac{d\tau}{dt}\right) = \frac{d\tau}{dt} L\left(\tau, x^j(\tau), \dot{x}^j\right) \tag{20}$$

Cancelling the scaling of the differential, we achieve the objective:

$$\int_{\tau_1}^{\tau_2} L\left(t(\tau), x^j(t(\tau)), \dot{x}^j \frac{d\tau}{dt}\right) \frac{dt}{d\tau} d\tau = \int_{\tau_1}^{\tau_2} L\left(\tau, x^j(\tau), \dot{x}^j\right) \frac{d\tau}{dt} \frac{dt}{d\tau} d\tau = \int_{\tau_1}^{\tau_2} L\left(\tau, x^j(\tau), \dot{x}^j\right) d\tau \tag{21}$$

We have replaced $t(\tau)$ with τ. The stretching or compressing of the clock parameter is canceled out via homogeneity of the Lagrangian function, keeping the action integral physically meaningful.

- **This is the big result that unlocks gauge invariance**. Just as we can choose the width of a rail line, we can choose how we measure time.

Since the action is a property of the path itself, which remains unchanged when we reparametrize it, values of this L cannot depend explicitly on the parameter. This first argument of L is only a label, a placeholder for whichever parameter is chosen, allowing both numerical and symbolic, formal equality: $L(t(\tau), \ldots) = L(\tau, \ldots)$.

Physical Interpretation

The physics is independent of the choice of parameter along the curve C traced out as the coordinates and velocities dynamically evolve. The Lagrangian scales exactly with the stretching or compressing of the parameter.

Decoupling Momenta and Velocities

Write the homogeneity of L (Equation 20) in the velocity argument more generally, removing unnecessary references to the parameter. Homogeneity is independent of choice of parameter; it is an intrinsic property of a parameter-invariant Lagrangian:

$$L\left(x^j, \lambda \dot{x}^j\right) = \lambda L\left(x^j, \dot{x}^j\right) \tag{22}$$

Differentiate both sides with respect to λ and set λ to 1:

$$\left.\frac{\partial L\left(x^j, \lambda \dot{x}^j\right)}{\partial \lambda}\right|_{\lambda=1} = \left.\frac{\partial L\left(x^j, \lambda \dot{x}^j\right)}{\partial \dot{x}^k} \dot{x}^k\right|_{\lambda=1} = \frac{\partial L}{\partial \dot{x}^k} \dot{x}^k = L\left(x^j, \dot{x}^j\right) \tag{23}$$

dropping the argument notation from L. Recognizing the conjugate momenta p_k, we get:

EULER'S IDENTITY

$$p_k \dot{x}^k = L \tag{24}$$

- *This is the identifying signal of a parameter-independent theory. It means that the Lagrangian is its own Legendre transform, and also that the Hamiltonian vanishes (see below)! Fear not, however, if you believe that the Hamiltonian is energy. Energy does not vanish, it just becomes the 0-th component of 4-momentum, as we see below. This is the first indication that we're going to get relativity "for free" as an unavoidable consequence of parameter invariance.*

Differentiate one more time with respect to $\dot{x}^j$, noting that $\frac{\partial \dot{x}^k}{\partial \dot{x}^j} = \delta_j{}^k$:

$$\frac{\partial L}{\partial \dot{x}^j} = \frac{\partial}{\partial \dot{x}^j}\left(\frac{\partial L}{\partial \dot{x}^k}\dot{x}^k\right) = \frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}\dot{x}^k + \frac{\partial L}{\partial \dot{x}^k}\frac{\partial \dot{x}^k}{\partial \dot{x}^j} = \frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}\dot{x}^k + \frac{\partial L}{\partial \dot{x}^j} \tag{25}$$

Implying

$$\frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}\dot{x}^k = 0 \tag{26}$$

or

$$\frac{\partial p_k}{\partial \dot{x}^j}\dot{x}^k = 0 \tag{27}$$

That $\frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}\dot{x}^k = 0$ means there is a non-zero vector $\dot{x}^k$ in the null space of the Hessian $\frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}$, further meaning that the Hessian is singular. By the inverse function theorem, we cannot solve the algebraic system $\frac{\partial p_k}{\partial \dot{x}^j}\dot{x}^k = 0$ for all velocities $\dot{x}^j$ in terms of the generalized momenta p_k.

- *Because the Hessian is singular, the* n *components of momentum are not algebraically independent; they are constrained to a surface of dimension* $n-1$ *or less. This is why the mass-shell condition* $p^2 = m^2$, redefining energy (!), *must exist—it is the defining condition of the narrow space into which the momenta have been squashed.*
- *The Hessian* $\frac{\partial^2 L}{\partial \dot{x}^j \partial \dot{x}^k}$ *is simply the Jacobian of the transformation from velocity-space to momentum-space, and that's the algebraic link.*
- *This failure is a feature, not a bug: it indicates redundant degrees of freedom, leading to gauge. The evolution of the unphysical components is arbitrary, corresponding to our freedom to choose a gauge.*

The Mass-Shell Condition and Vanishing Hamiltonian

Because the mapping from velocity space to momentum space is non-invertible, the momenta p_k are not algebraically independent of one another. They are geometrically squashed onto a lower-dimensional surface within the cotangent space. This restriction manifests as an *algebraic constraint* called the ***Primary Constraint***:

$$\phi(p_k, x^k) = 0 \tag{28}$$

The Leap to Special Relativity

Consider the Lagrangian for a relativistic free particle of mass m. This is the canonical example of a parameter-invariant (homogenous-degree-1) system:

$$L = m\sqrt{\eta_{\mu\nu}\,\dot{x}^{\mu}\,\dot{x}^{\nu}} \tag{29}$$

where $\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$, the Minkowski metric.

- *Up to this point, our discussion of the manifold X_n has been purely topological—we have not described how to measure distance or "length" along a path. A realistic Lagrangian of a free particle should depend only on its velocities, just as in Classical Mechanics. If we further require parameter-invariance, so that the geometric history of the particle doesn't depend on our choice of clock, we must find a Lagrangian that is homogeneous of degree 1 in the velocities. The most natural candidate for such a function is the arc-length of the curve .*
- *Turns out this is also the optimal candidate, by some definition of "optimal."*
- *To define "length" when Newtonian time is demoted to an ordinary coordinate, we introduce a **metric tensor**. In the flat space-time of Special Relativity, that's the Minkowski metric, the "ruler" for measuring proper distance in the manifold. It lets the Lagrangian also be a Lorentz-invariant scalar quantity. By defining the Lagrangian as the square root of the squared magnitude of the velocity vector, we satisfy two requirements at once: (1) the action is computed from geometrical intervals along the path by the **proper time**, and (2) square root structure ensures homogeneity of degree 1.*
- *This leap into Special Relativity doesn't just change formulas. We are choosing the specific geometry that allows our parameter-invariant machinery to produce the energy-momentum relations of the real world.*

The conjugate momenta, $p_\mu = \frac{\partial L}{\partial x^\mu}$, are

INDEX LOWERING

$$p_\mu = \frac{m\,\dot{x}_\mu}{\sqrt{\dot{x}_\alpha\,\dot{x}^\alpha}} \tag{30}$$

- *In Equation 30, we transitioned the "up" indices of $\eta_{\mu\nu}\,\dot{x}^{\mu}\,\dot{x}^{\nu}$ to one down and one up, $\dot{x}_\alpha\,\dot{x}^\alpha$*, absorbing *$\eta_{\mu\nu}$ in the offing, with renaming of the dummy indices. Of course, both expressions suffer the summation convention over repeated down-up pairs. In tensor calculus, this is not a mere change of notation, but a geometric operation: **index raising or lowering.** Recall that velocity $\dot{x}^\mu$ is a contravariant vector in the tangent space. However, conjugate momentum p_μ is a covariant covector in the cotangent space. To relate the two, we need a bridge between the tangent space and the cotangent space. The metric tensor* $\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$ *is that bridge.*
- *We define the covariant components of the velocity by contracting the contravariant components with the metric: $\dot{x}_\mu = \eta_{\mu\nu}\,\dot{x}^\nu$. Physically, this operation uses the local geometry to "measure" the vector and turn it into a linear function of any vector—**that's the formal definition of a covector**. The sigil of this linear function is the down index, just waiting to be contracted against the up index of a vector. Contraction performs the linear operation and produces a scalar result. This is just a fancy inner product or dot product that works with any kind of coordinate system with a metric. The familiar dot product of two Cartesian vectors works only in Cartesian spaces, where we don't distinguish vectors and covectors.*

- *In the relativistic Lagrangian* $L = m\sqrt{\eta_{\mu\nu}\,\dot{x}^{\mu}\,\dot{x}^{\nu}} = m\sqrt{\dot{x}_{\nu}\,\dot{x}^{\nu}}$ *, the metric is baked into the arc-length.*
- *This "sleight-of-hand" is the core of the Hamiltonian and Lagrangian treatments: the metric lets us swap between the "velocity" description of a path and the "momentum" description of a state, ensuring that the final constraint* $p_{\mu}\,p^{\mu} \equiv p^2 = m^2$ *is a true scalar invariant, independent of coordinate system.*
- *This tangent-cotangent, vector-covector duality and all these formulas carry over to General Relativity.*
- *We can now conflate overdot and overtick because we assume parameter-invariance.*

If we square the momentum, the velocity terms cancel, leaving us with a pure identity for the momenta:

$$p_{\mu}\,p^{\mu} = \frac{m^2\,\dot{x}_{\mu}\,\dot{x}^{\mu}}{\left(\sqrt{\dot{x}_{\alpha}\,\dot{x}^{\alpha}}\right)^2} = m^2 \tag{31}$$

This is often quoted as Einstein's most famous formula, (in natural units, where the speed of light $c = 1$.

$E^2 - \mathbb{p}\bullet\mathbb{p} = m^2$

where $E = p_0$, the 0 component of p_{μ}, is the new home for energy, and $\mathbb{p}$ is the space part of the 4-momentum p_{μ}, our old friends, the conjugate momenta.

- *This is also called* ***the mass-shell constraint****, with "shell" meaning "pertaining to classically realizable motions, relativistic but non-quantum." In quantum physics, we must consider off-shell motions, because a quantum system "considers all paths" between events.*

Einstein's formula is a necessary consequence of the singular Hessian, not a magically discovered postulate.

- *The particular choice of* $\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$ *forces the minus sign in* $E^2 - \mathbb{p}\bullet\mathbb{p} = m^2$*. Other authors quote* $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$*, which generates the same physics*.

The Vanishing Hamiltonian

We already noted in Euler's Identity, Equation 24, that $p_{\mu}\,\dot{x}^{\mu} = L$. But the definition of the Hamiltonian is

$$H = p_{\mu}\,\dot{x}^{\mu} - L \tag{32}$$

That's now 0! This means that the classical "energy" of the system, represented by the Hamiltonian, vanishes identically. However, just as t has been *demoted* to a mere coordinate, energy has been *promoted* to the 0 component of the 4-momentum.

The deeper meaning, though, is that in a gauge-invariant theory, the Newtonian time coordinate t does not generate dynamical evolution. **The entire physics is contained in the constraints!**

Although this might seem shocking, it is 100% in keeping with the original ideas of d'Alembert and Lagrange: "capture physics by capturing constraints." It's sobering to realize that Special Relativity was hiding in plain sight in the old Lagrangian and Hamiltonian, long before Einstein discovered it.

Roadmap to Quantum Fields

We have reached the following conclusion: when we insist that physics be independent of both the coordinate system and the parameter “clock,” we are rewarded with the mass-shell constraint $p^2 = m^2$. This is the end of our non-quantum road, but the beginning of the road to the Standard Model.

Lots of detail is swept under the rug in the following sketches, but the roadmap forward should be clear.

Quantization: Constraints as Wave Equations

In the non-quantum Hamiltonian approach, we promote momentum variables to operators ($p \to \hat{p}$, see Goldstein). In a non-singular system, we would plug these into the Hamiltonian to find the energy. But our constrained Hamiltonian vanishes!

Our physics is already relativistic, so, to quantize, we just follow the Dirac Prescription for constrained systems: the physical quantum states $\left|\psi\right\rangle$ must be annihilated by the constraint operator:

$$\hat{\phi}\left|\psi\right\rangle = 0 \Longleftrightarrow \left(p^2 - m^2\right)\left|\psi\right\rangle = 0 \tag{33}$$

By substituting the quantum momentum operator $p_\mu = -i\hbar\, \frac{\partial}{\partial x^\mu}$, we get the relativistic Klein-Gordon Equation:

$$\left(\Box + m^2\right)\left|\psi\right\rangle \equiv \left(\frac{\partial^2}{\partial t^2} - \nabla^2\right)\left|\psi\right\rangle = 0 \tag{34}$$

with $c = 1$, $\hbar = 1$ everywhere. The “unsolvability” of the non-quantum momenta produces to one of the wave equations of relativistic quantum mechanics!

From Particles to Fields

To reach the Standard Model, we transition from the world-line of a single particle to a field-dependent Lagrangian Density $\mathcal{L}(\phi, \partial_\mu \phi)$.

Local Gauge Invariance

Just as we required invariance under $t \to \tau$ for a particle, we require field theories to be invariant under other kinds of gauge transformations, e.g., under local phase shifts $\phi(x) \to e^{i\alpha(x)}\, \phi(x)$, where α varies smoothly with x.

The Return of the Singularity

Just as with our relativistic particle, the Lagrangian (densities) for the photon (Maxwell), the weak-force bosons $W^\pm$, Z (Weinberg-Salam), and gluons (Yang-Mills) have singular Hessians. They are gauge theories precisely because they possess this mathematical redundancy.

The Ultimate Objective: The Standard Model

By following this exact procedure—identifying the symmetries, finding the singular Hessians, and enforcing the resulting constraints—we derive the quantum-theoretical forces of nature, except for gravitation:

Electromagnetism: from U(1) gauge invariance.

The Weak Force: born from SU(2) gauge invariance.

The Strong Force (QCD): born from SU(3) gauge invariance.

The Outlier: General Relativity

Differential geometry first came into physics with General Relativity and gravitation. It was only discovered much later that the same mathematics underlies quantum field theory. Gravitation has its revenge, however: no one knows how to fit gravitation into the same scheme as the other quantum fields. It’s even possible to do quantum field theory on top of general relativity, in a highly curved space-time, but the gauge theories themselves enjoy their own differential geometries. It’s as though the gods and the mortals have a common ancestor, who is laughing at their inability to get along.

Conclusion

The “magical Lagrangian” is more than a tool for solving pulleys and pendulums. It is a dictionary that translates the geometry of a manifold into the laws of the Universe. By mastering the singular Hessian and the covariant covector, we have learned the language in which the Standard Model is written.

Part II: $\mathcal{UD}$ Tensorial Calculus

Abstract

In this notebook, I propose a technique in the Wolfram language, $\mathcal{UD}$ (up-down) indexing, for tensorial calculations in physics. The technique is self-contained, has no external dependencies, and is novel, so far as I know. The implementation is an early prototype of an embedded compiler for a $\mathcal{UD}$ calculus. Several directions for improvement are noted as we go through examples.

In Part I, I showed how tensorial forms from Lagrangians not only account for relativity, but *force* it. To keep that part short, it contained only informal mathematics. I saved Wolfram code—formal mathematics—for this part. Here we exhibit $\mathcal{UD}$ indexing on a specific example: *relativistic aberration*.

Carlip’s paper, Aberration and the Speed of Gravity starts with electrodynamical aberration, then progresses to the gravitational case. We formalize and check his math.

Introduction: Aberration

Carlip’s paper resolves a conundrum in gravitation. No causal signal can travel faster than *c*, the speed of light in the vacuum, measured in any reference frame. But, if you consider, say, the Earth-Sun system, and then you point the force vector from the Earth to the ***retarded position*** of the Sun, where it was “eight minutes ago,” when any gravitational signals were emitted, your calculation of Earth’s orbit will be wrong by a wide margin. This “miss” happens both in Newtonian and general-relativistic gravitation. You must point the force vector roughly at the “instantaneous,” current position of the Sun, as measured in some space-time coordinate system at rest with the Sun. This doesn’t sound right! What is going on?

- *Actually, you must point the force vector roughly at the instantaneous center of gravity of the Earth-Sun system, which is almost at the center of the Sun. Such is likely a negligible quibble compared to aberration, magnetohydrodynamical fluctuations moving mass around, etc.*
- *In the gravitational context, this effect is an instance of* astronomical aberration.

Electrodynamical Aberration

It turns out that the same thing happens in electrodynamics, which has a central-force law similar to gravitation’s. The electric field does not point to the retarded position of the source!

Electrodynamics is easier to analyze because we don’t need Riemann curvature tensors and Christoffel symbols. Flat Minkowski space-time suffices for initial demonstration of $\mathcal{UD}$ indexing.

Problem Statement

Consider a charged source particle, $z(s)$, moving along some timelike curve parameterized by s. Now consider a stationary test particle, x, at some spacelike distance from z and ask “what is the 3D direction of the electric field at x?”

- *We’ll find that it points (roughly) toward the instantaneous position of z, that is, where z is “now” in the lab frame, rather than toward its* ***retarded position*** *$z(s_R)$, where z was some time ago. “Roughly” means “to first order in the velocity of the source,” or, “assuming the source doesn’t accelerate.”*
- *The apparent location of the source amounts to a linear extrapolation from the retarded position of the source—where it was—forward by the source’s retarded velocity—the speed and direction when and where it was.*
- *This “conspiracy” might lure one into thinking that the signal propagates faster than is possible, faster than c. However, all the expressions depend only on information at the retarded time, so relativity is still safe and true! Carlip explains more deeply in his Section 3. We’re just checking his math via the Wolfram language and our $\mathcal{UD}$ calculus..*

Valence and $\mathcal{UD}$ Indexing

Let the ***valence*** of a tensorial expression track the up-ness and down-ness of its indices. λ^μ has valence $\mathcal{U}$, σ_μ has valence $\mathcal{D}$, $\eta_{\mu\nu}$ has valence $\mathcal{D}^2$, $\eta^\nu{}_\mu$ has valence $\mathcal{UD}$, and so on.

To specify indices precisely in input, writeλ^μ as `λ[𝒰[μ]]`, σ_μ as `σ[𝒟[μ]]`, $\eta_{\mu\nu}$ as `η[𝒟[μ],𝒟[ν]]`, $\eta^\nu{}_\mu$ as `η[𝒰[ν],𝒟[μ]]`, and so on. Display rules (below) serve to show such expressions in a human-readable form in output boxes.

The Summation Convention

Einstein’s summation convention for contraction is over repeated up-down index pairs in products like $\lambda^\mu\,\sigma_\mu$ and tensorial expressions like $\eta_\mu{}^\mu$. This convention obtains everywhere in this notebook and will be invoked without mention.

For this notebook, one example is especially important:
$\eta_{\mu\nu}\,\sigma^\mu = \sigma_\nu = (\sigma_0, \sigma_1, \sigma_2, \sigma_3) = \left(\sigma^0, -\sigma^1, -\sigma^2, -\sigma^3\right)$, where $\eta_{\mu\nu}$, the Minkowski metric, is diag(1, −1, −1, −1). $\eta_{\mu\nu}$ transforms the contravariant components $\left(\sigma^0, \sigma^1, \sigma^2, \sigma^3\right)$ into the covariant components $(\sigma_0, \sigma_1, \sigma_2, \sigma_3)$, which numerically equal $\left(\sigma^0, -\sigma^1, -\sigma^2, -\sigma^3\right)$.

Display Rules

To make results easier for humans to read, apply these rules to final results. The output will no longer be recognizable to our tools, so apply these rules only as a final step, “exiting” a calculation.

```
In[1]:= ClearAll[dpyRules, d, s, λ, σ, η, dλds];
dpyRules = {
    (*Specific rules ahead of general rules*)
    dλds[𝒟[ν_]] :→ (FractionBox[d Subscript[λ, ν], d s] // DisplayForm),
    dλds[𝒰[ν_]] :→ (FractionBox[d Superscript[λ, ν], d s] // DisplayForm),
    (*General Rules*)
    ε_[𝒰[τ_]] :→ Superscript[ε, τ],
    ε_[𝒟[τ_]] :→ Subscript[ε, τ],
    ε_[𝒟[μ_], 𝒟[ν_]] :→ Subscript[ε, (μ ν)],
    ε_[𝒟[μ_], 𝒰[ν_]] :→ Superscript[Subscript[ε, μ], ν],
    ε_[𝒰[μ_], 𝒰[ν_]] :→ Superscript[ε, (μ ν)],
    ε_[𝒰[μ_], 𝒟[ν_]] :→ Subscript[Superscript[ε, μ], ν]
   };
ClearAll[echo];(*for debugging*)
echo[τ_] := Module[{}, Print[τ /. dpyRules]; Echo[τ]];
(* unit tests *)
(σ[𝒟[ν]] == η[𝒟[μ], 𝒟[ν]] × σ[𝒰[μ]]) /. dpyRules
dλds[𝒰[ν]] /. dpyRules
```

Out[5]= $\sigma_\nu == \eta_{\mu\,\nu}\,\sigma^\mu$

Out[6]//DisplayForm=
$$\frac{d\,\lambda^\nu}{d\,s}$$

Computational Plan: Gradients

- *This notebook assumes natural units, where $c = 1$, velocities are dimensionless, and space and time are commensurable. We do not perform detailed dimensional analysis (TODO: that might be an interesting thing to do).*

The plan is to get the electric field at x from gradients of the Liénard-Wiechert potential.

Liénard-Wiechert Potential

Here is Carlip's definition, in covariant form:

In[7]:=
```
A[𝒟[ν_]] := e/r λ[𝒟[ν]];
A[𝒟[ν]] /. dpyRules
```

Out[8]= $\frac{e\,\lambda_\nu}{r}$

where

- e is a constant electric charge
- r is the ***retarded separation*** distance in the lab frame (yet to be calculated)
- λ_ν is the ***retarded velocity*** $dz(s_R)_\nu / ds$, again in covariant form
  - *It's decidedly NOT $\partial z_\nu / \partial x^\mu$, which doesn't even have the right valence.*

Field Strength Faraday Tensor

The field gradients are

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_n A_\mu \tag{35}$$

a textbook formula.

To compute $F_{\mu\nu}$, we'll need gradients for r and λ because of the chain rule.

Begin with the ***separation vector*** $\sigma^\mu = (x^\mu - z^\mu(s_R))$ between the emission ***event*** at the retarded 4-position $z^\mu(s_R)$ and the absorption event at 4-position x^μ. σ^μ depends on x implicitly through s_R. So we'll start with the gradient $\partial s_R / \partial x^\mu \equiv \partial_\mu s_R$.

$\partial_\mu s_R$ from the Null Interval

The only leverage we have on s_R is through the length of the separation vector, the ***proper time*** $\sigma^\mu\,\sigma_\mu$. It is zero, a ***null interval***, as with all light-like paths (on the light cone) in a vacuum:

THE NULL INTERVAL

$$\phi\,(x, s_R) \equiv \sigma_\nu\,\sigma^\nu = \eta_{\mu\nu}(x^\mu - z^\mu(s_R))\,(x^\nu - z^\nu(s_R)) = 0 \tag{36}$$

This is Equation 1.1 in Carlip's paper.

Abbreviate s_R to s

For now, write s for s_R to save space and to avoid confusing subscript R (retarded) with tensorial subscripts. Just remember that $s_R \to s$ is a specific, dynamical variable that depends on x. It is also a specific value of s as a generic path parameter of z, but that fact does not concern us.

In[9]:=
```
ClearAll[η, ϕ, σ, λ, r, e, x, z, s];
```

Here is a good-looking Wolfram version of Equation 36:

In[10]:=
```
(ϕ = (η[𝒟[μ], 𝒟[ν]] (x[𝒰[μ]] - z[s][𝒰[μ]]) (x[𝒰[ν]] - z[s][𝒰[ν]]))) /. dpyRules
```

Out[10]=

$\eta_{\mu\nu} (x^\mu - z[s]^\mu) (x^\nu - z[s]^\nu)$

$\partial_\mu s$ Via the Implicit Function Theorem

The fact that $\phi(x, s) = 0$ means that ϕ qualifies for the implicit function theorem. By the chain rule, the gradient $\partial_\mu \phi$ is

$$\partial_\mu \phi + \frac{d\phi}{ds} \partial_\mu s = 0 \tag{37}$$

Thus, the gradient we seek is $\partial_\mu s = -\partial_\mu \phi / \frac{d\phi}{ds}$, where $\partial_\mu \phi$ is a 4-vector and $\frac{d\phi}{ds}$ is a scalar. Let's compute the factors $\partial_\mu \phi$ and $\frac{d\phi}{ds}$ one at a time.

4-Vector $\partial_\mu \phi$

The standard derivative operator `Dt` is agnostic to valence and cannot apply metrical contractions. Applying `Dt` blindly produces chained derivatives of indices and the $\mathcal{UD}$ symbols, which don't make sense. Therefore, we just explicitly define dϕds

In[11]:=
```
(dϕdx = 2 σ[𝒟[μ]]) /. dpyRules
```

Out[11]=

$2\,\sigma_\mu$

Here is the derivation:

$$\begin{aligned}
&\frac{\partial\phi}{\partial x^\rho}[\eta_{\mu\nu}(x^\mu - z^\mu(s_R))\,(x^\nu - z^\nu(s_R))] \\
&= \eta_{\mu\nu}(x^\nu - z(s)^\nu)\,\frac{\partial x^\mu}{\partial x^\rho} + \eta_{\mu\nu}(x^\mu - z(s)^\mu)\,\frac{\partial x^\nu}{\partial x^\rho} \\
&= 2\,\eta_{\mu\rho}(x^\mu - z^\mu(s)) \\
&= 2\,\eta_{\mu\rho}\,\sigma^\mu \\
&= 2\,\sigma_\rho \\
&= 2\left(\sigma^0, -\sigma^1, -\sigma^2, -\sigma^3\right)
\end{aligned} \tag{38}$$

invoking symmetry of η.

Scalar $d\phi/ds$

This time, we *do* want `Dt`'s chain rule:

In[12]:=
```
ClearAll[η];
(dϕds = D[ϕ, s]) /. dpyRules
```

Out[13]=

$$-\eta_{\mu\,\nu}\,(x^{\nu} - z[s]^{\nu})\,z'[s]^{\mu} - \eta_{\mu\,\nu}\,(x^{\mu} - z[s]^{\mu})\,z'[s]^{\nu}$$

This is pretty clearly $-2\,\sigma_{\mu}\,\lambda^{\mu}$ because $\lambda \equiv z'(s)$, but the display is not good enough. Now is the time to introduce ***tensor rules***, a combination of transformations of our special tensors and transformations for all tensors:

Tensor Rules

In[14]:=
```
ClearAll[tensorRules];
tensorRules = {
   (*Special Rules First*)
   (*Definition of σ*)
   x[ud_] - z[s][ud_] :→ σ[ud],
   (*Null Interval*)
   σ[𝒰[α_]] × σ[𝒟[α_]] :→ 0,
   σ[𝒟[α_]] × σ[𝒰[α_]] :→ 0,
   (*Definition of λ*)
   z'[s][ud_] :→ λ[ud],
   (*Unit Velocity*)
   λ[𝒰[α_]] × λ[𝒟[α_]] :→ 1,
   λ[𝒟[α_]] × λ[𝒰[α_]] :→ 1,
   (*Mixed Metric η_μ^ρ acts as δ_μ^ρ*)
   η[𝒟[μ_], 𝒰[ρ_]] :→ δ[𝒟[μ], 𝒰[ρ]],
   (*Metric Symmetry, avoiding unbounded recursion via OrderedQ*)
   η[𝒟[α_], 𝒟[β_]] /; OrderedQ[{α, β}] :→ η[𝒟[β], 𝒟[α]],
   (*General Rules*)
   (*Coalesce Terms into Down-Up Order*)
   τ_[𝒟[μ_]] × υ_[𝒰[μ]] /; OrderedQ[{τ, υ}] :→ υ[𝒟[μ]] × τ[𝒰[μ]],
   (*Raising and Lowering: e.g. η_μα λ^α→λ_μ*)
   η[𝒟[μ_], 𝒟[ν_]] × τ_[𝒰[ν_]] :→ τ[𝒟[μ]],
   η[𝒟[μ_], 𝒟[ν_]] × τ_[𝒰[μ_]] :→ τ[𝒟[ν]],
   η[𝒟[μ_], 𝒟[ν_]] (τ_ + υ_) :→ η[𝒟[μ], 𝒟[ν]] τ + η[𝒟[μ], 𝒟[ν]] υ,
   η[𝒟[μ_], 𝒟[ν_]] (τ_[𝒰[ν_]] × υ_[𝒰[μ_]]) :→ τ[𝒟[μ]] × υ[𝒰[μ]],
   (*unraveling recursive 𝒰𝒟 indices*)
   τ_[(h1 : 𝒰 | 𝒟)[(h2 : 𝒰 | 𝒟)[ud_]]] /; h1 === h2 :→ τ[h1[ud]]
  };
```

- *This is just a beginning. By extending the tensor rules to more elaborate examples we could eventually have an embedded compiler for the 𝒰𝒟 calculus.*
- *The unit-velocity rule follows from the fact that if we parametrize a curve by its arc length, its tangent vector is always of norm 1. This is true in Minkowski space, too, because*

$$ds^2 = \eta_{\mu\nu}\, dz^\mu\, dz^\nu \tag{39}$$

$$1 = \eta_{\mu\nu} \frac{dz^\mu}{ds} \frac{dz^\nu}{ds} \tag{40}$$

NORM OF LAMBDA

$$1 = \eta_{\mu\nu}\, \lambda^\mu\, \lambda^\nu = \lambda_\nu\, \lambda^\nu \tag{41}$$

Now we can see **dϕds** more clearly

In[16]:=
```
(dϕds = (dϕds //. tensorRules)) /. dpyRules
```
Out[16]=

$$-\sigma_\mu\, \lambda^\mu - \lambda_\nu\, \sigma^\nu$$

Invariant Retarded Distance *r*

The gradient $\partial_\mu s$ is the negative ratio of these two factors: **-dϕdx/dϕds**. If we write the ratio, however, Wolfram will cheerfully cancel σ_μ. Wolfram doesn't "know" that μ in $\sigma_\mu\, \lambda^\mu$ is a "dummy" index and must not be allowed to collide with the μ in σ_μ in the numerator. We might replace $\sigma_\mu\, \lambda^\mu$ with $\sigma_\rho\, \lambda^\rho$ via a new tensor rule that checks for collisions.

- *TODO: Avoiding name collisions in general is a deep topic in term-rewriting, the underlying evaluation theory of Wolfram. Implementing it for 𝒰𝒟 indexing is a potentially interesting topic, taking us deeper into compiler territory. It's too much, now, for this prototype. Likewise for coalescing dummy indices, i.e., detecting that $\sigma_\mu\, \lambda^\mu = \sigma_\rho\, \lambda^\rho$.*

A better idea is to follow Carlip's Equations 1.3 and 1.4 and simply define $\sigma_\mu\, \lambda^\mu$ as the scalar ***invariant retarded distance*** *r* via some new scalar rules:

Scalar Rules

In[17]:=
```
ClearAll[scalarRules];
scalarRules = {
   σ[(h1 : 𝒰 | 𝒟) [ud_]] × λ[(h2 : 𝒰 | 𝒟) [ud_]] /; h1 =!= h2 :→ r
  };
```

- *This rule illustrates a complex pattern that matches product expressions containing tensors with opposite parities, subject to the summation convention. It will match either $\sigma^\mu\, \lambda_\mu$ or $\sigma_\mu\, \lambda^\mu$. It's debatable whether that pattern is more readable than a pair of explicit patterns,* `σ[𝒰[μ_]]λ[𝒟[μ_]]:>r, σ[𝒟[μ]]λ[𝒰[μ]:>r`*, as in* `tensorRules`*. I have illustrated both styles. Which you choose is a matter of taste.*

In[19]:=
```
dϕds /. scalarRules /. dpyRules
```
Out[19]=

$$-2\, r$$

In[20]:=
```
(gradS = dϕdx / (dϕds /. scalarRules)) /. dpyRules
```
Out[20]=

$$-\frac{\sigma_\mu}{r}$$

Again, this is manifestly covariant, as all gradients should be.

Writing out Carlip's Equations 1.3 and 1.4 consistently with the above:

GRADIENT OF SR

$$\partial_\mu s_R(x) = \frac{\sigma_\mu}{r(x)} \tag{42}$$

INVARIANT RETARDED DISTANCE

$$r(x) = \lambda^\mu (s_R)\, \sigma_\mu \tag{43}$$

r has the velocity-dependent terms that are the key to the mystery of aberration.

Let's capture r for future use:

In[21]:=
```
ClearAll[rExpr];
(rExpr = σ[𝒟[ρ]] × λ[𝒰[ρ]]) /. dpyRules
```

Out[22]=

$\sigma_\rho\, \lambda^\rho$

Field Strength Faraday Tensor

To get the field-strength Faraday tensor, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, we'll need gradients for e, r, σ, and λ as we grind through the chain rule applied to A.

In[23]:=
```
?? A
```

Out[23]=

Symbol

Global`A

Definitions

A[𝒟[ν_]] := $\frac{e\, \lambda[\mathcal{D}[\nu]]}{r}$

Full Name Global`A

Let's write some specialized rules in a custom 4-grad operator to get us through the computation.

Custom 4-grad Operator

Clear out residual definitions, invalidating earlier expressions in the notebook that depend on them.

In[24]:=
```
ClearAll[fourGrad, σ, λ, η, r, e, dλds, δ, F];
```

Basic Rules from Calculus

Linearity, Product Rule, Power Rule

In[25]:=
```
fourGrad[a_ + b_, ud_] := fourGrad[a, ud] + fourGrad[b, ud];
fourGrad[a_ * b_, ud_] := b fourGrad[a, ud] + a fourGrad[b, ud];
fourGrad[a_^n_, ud_] := n a^(n-1) fourGrad[a, ud];
```

Constant *e*

In[28]:=
```
fourGrad[e, _] := 0;
```

Gradient of the Separation Vector σ

We'll need $\partial_\mu \sigma^\nu$:

$$\partial_\mu \sigma^\nu = \partial_\mu (x^\nu - z^\nu(s_R)) = \partial_\mu x^\nu - \partial_\mu z^\nu(s_R) = \delta_\mu^\nu - \frac{dz^\nu}{ds_R} \partial_\mu s_R \tag{44}$$

The first term, the Kronecker delta, in Lorentz-invariant form, is the Minkowski metric η.

For the second term, we've already defined $\lambda^\nu \equiv \frac{dz^\nu}{ds_R}$, and we've already calculated $\partial_\mu s_R = \sigma_\mu / r$ (Equation 42).

The following works whether μ is up or down, ditto ν:

In[29]:=
```
fourGrad[σ[udν_], udμ_] := η[udμ, udν] - λ[udν] σ[udμ]/r;
```

This is the one we'll need

In[30]:=
```
fourGrad[σ[𝒟[ν]], 𝒟[μ]] /. dpyRules
```

Out[30]=

$$\eta_{\mu\,\nu} - \frac{\lambda_\nu\,\sigma_\mu}{r}$$

Gradient of the Velocity λ

Now follows the gradient of λ:

$$\partial_\mu \lambda_\nu \equiv \frac{d\lambda_\nu(s_R)}{dx^\mu} \tag{45}$$

Remembering that λ depends only implicitly on x:

$$\frac{d\lambda_\nu(s_R)}{dx^\mu} = \left(\frac{\partial \lambda_\nu(s_R)}{\partial x^\mu} = 0\right) + \frac{d\lambda_\nu(s_R)}{ds_R} \partial_\mu s_R \tag{46}$$

Let's just define the symbol **dλds** for $\frac{d\lambda_\nu(s_R)}{ds_R}$ (we don't need to decompose or evaluate it to get Carlip's Equation 1.7), and we already know $\partial_\mu s_R = \sigma_\mu / r$:

In[31]:=
```
fourGrad[λ[udν_], udμ_] := dλds[udν] σ[udμ]/r;
```

In[32]:=
```
fourGrad[λ[𝒟[ν]], 𝒟[μ]] /. dpyRules
```

Out[32]=

$$\frac{\frac{d\,\lambda_\nu}{d\,s}\,\sigma_\mu}{r}$$

$\partial_\mu r$: Gradient of r

Reminder that r is a contraction:

In[33]:=
```
rExpr /. dpyRules
```

Out[33]=

$$\sigma_\rho \lambda^\rho$$

Our fourGrad has the product rule and the special rules above for gradients of σ and λ. Start with a "raw" expression for $\partial_\mu r$:

In[34]:=
```
(gradRRaw = fourGrad[rExpr, 𝒟[μ]]) /. dpyRules
```

Out[34]=

$$\frac{\frac{d\lambda^\rho}{ds}\sigma_\mu \sigma_\rho}{r} + \left(\eta_{\mu\rho} - \frac{\lambda_\rho \sigma_\mu}{r}\right)\lambda^\rho$$

Now "cook" the "raw" expression for $\partial_\mu r$ and "bake" it into a "frozen" rule for **fourGrad**:

In[35]:=
```
(fourGrad[r, μ_] = Simplify[gradRRaw //. tensorRules] //. tensorRules) /. dpyRules
```

Out[35]=

$$\lambda_\mu + \frac{\sigma_\mu \left(-1 + \frac{d\lambda^\rho}{ds}\sigma_\rho\right)}{r}$$

Grinding through textbook definitions, with a final step to Collect terms with common powers of r and e:

In[36]:=
```
Fraw = (fourGrad[A[𝒟[ν]], 𝒟[μ]] - fourGrad[A[𝒟[μ]], 𝒟[ν]]) //. tensorRules;
```

In[37]:=
```
Collect[Fraw, {r, e}, FullSimplify] /. dpyRules
```

Out[37]=

$$\frac{e\left(\frac{d\lambda_\nu}{ds}\sigma_\mu - \frac{d\lambda_\mu}{ds}\sigma_\nu\right)}{r^2} + \frac{e\left(-\lambda_\nu \sigma_\mu + \lambda_\mu \sigma_\nu\right)\left(-1 + \frac{d\lambda^\rho}{ds}\sigma_\rho\right)}{r^3}$$

This reproduces Carlip's Equation 1.7 closely, and exactly after multiplying -1 through each of the parenthesized factors in the $1/r^3$ term.

Resolving Aberration

We close by deriving Carlip's Equations 1.8 and 1.9 and recapitulating his argument that the electric field, while incorporating only retarded terms, thus relativistically correct, "bends" the force vector via its velocity-dependent terms, toward the "instantaneous" direction. If the source does not change its velocity, the correction is exact. Relativity maintains the illusion of pointing the force, measured at the test location x, towards the instantaneous position of the source.

Start by decomposing the separation and velocity into their space and time components. We're interested specifically in the space components to assess the direction in which they point.

Components of the Separation Vector

Start again with the definition

$$\sigma^{\mu} = x^{\mu} - z^{\mu}(s_R) \tag{47}$$

which has space (3-vector) slice

$$\sigma^{\text{space}} = \boldsymbol{x} - \boldsymbol{z}(s_R) \tag{48}$$

defining 3-vectors $\boldsymbol{x}$ and $\boldsymbol{z}(s_R)$ *en passant*. The ***retarded spatial distance*** R, not invariant like r, is

$$R \equiv |\boldsymbol{x} - \boldsymbol{z}(s_R)| \tag{49}$$

Defining a unit vector pointing from $\boldsymbol{x}$ to the retarded position of the source, $\boldsymbol{z}(s_R)$

$$\hat{\boldsymbol{n}} = \frac{\boldsymbol{x} - \boldsymbol{z}(s_R)}{R} \tag{50}$$

we get

SEPARATION VECTOR COMPONENTS

$$\sigma^{\text{space}} = R\hat{\boldsymbol{n}} \tag{51}$$

Because $\left(\sigma^0\right) = |\boldsymbol{x} - \boldsymbol{z}(s_R)|^2$, we conclude that

$$\sigma^0 = R \tag{52}$$

Velocity-Dependent Terms

The velocity 4-vector, tangent to the curve at event $z(s_R)$, is

VELOCITY 4-VECTOR

$$\lambda^{\mu} = \frac{dz}{ds} = \frac{dz}{dt}\frac{dt}{ds} = \frac{d}{dt}(t, \boldsymbol{z}(\boldsymbol{s}))\left.\frac{dt}{ds}\right|_{s=s_R} = (1, \boldsymbol{v}_R)\left.\frac{dt}{ds}\right|_{s=s_R} \tag{53}$$

Because

$$ds^2 = dt^2 - d\boldsymbol{z}^2 \tag{54}$$

$$1 = \left(\frac{dt}{ds}\right)^2 - \left(\frac{d\boldsymbol{z}}{dt}\right)^2\left(\frac{dt}{ds}\right)^2 \tag{55}$$

we can solve for dt/ds:

In[38]:=
```
Solve[1 == dtds^2 - dzdt^2 dtds^2, dtds]
```

Out[38]=

$$\left\{\left\{\text{dtds} \to -\frac{1}{\sqrt{1 - \text{dzdt}^2}}\right\}, \left\{\text{dtds} \to \frac{1}{\sqrt{1 - \text{dzdt}^2}}\right\}\right\}$$

But we defined $\boldsymbol{v}_R = \left.\frac{d\boldsymbol{z}}{dt}\right|_{s=s_R}$. Substitute that inside the radical, and choose the positive solution for a time-forward path:

$$\left.\frac{dt}{ds}\right|_{s=s_R} = \frac{1}{\sqrt{1 - \boldsymbol{v}_R{}^2}} \equiv \gamma_R \tag{56}$$

Rewrite the velocity four-vector of the source, Equation 53:

$$\lambda^{\mu} = \gamma_R(1, \boldsymbol{v}_R) \tag{57}$$

so that

VELOCITY COMPONENTS

$$\lambda^0 = \gamma_R \tag{58}$$

$$\lambda^i = \gamma_R \boldsymbol{v}_R{}^i \tag{59}$$

with the Latin index i iterating over the spatial components, as usual.

The conclusions above constitute Equation 1.5 in Carlip's paper. In summary, R is the retarded spatial distance from $z(s_R)$ to x, $\boldsymbol{v}_R$ is the retarded 3-velocity, and the unit 3-vector $\hat{\boldsymbol{n}}$ points to the retarded position. Nothing here hints at anything but stock special relativity and electrodynamics. But does $\hat{\boldsymbol{n}}$ point in the direction of the force vector? To find out, **we must fish out the electric field from the field strength tensor**.

Break up the invariant retarded distance, r, referring to its definition, Equation 43:

$$r = \lambda^0\, \sigma_0 + \lambda^i\, \sigma_i \tag{60}$$

where Latin indices, like i, sum over the spatial components 1, 2, 3.

From the velocity components (Equations 58, ff), we get

$$r = \gamma_R\, \sigma^0 + \gamma_R\, \boldsymbol{v}_R{}^i\, \sigma_i \tag{61}$$

in terms of the retarded 3-velocity, $\boldsymbol{v}_R{}^i$. Now substitute components of the separation vector σ from Equations 51, ff

SPACE COMPONENTS OF INVARIANT RETARDED DISTANCE

$$r = R\gamma_R - R\gamma_R\, \boldsymbol{v}_R{}^i\, \hat{\boldsymbol{n}}_i = R\gamma_R\left(1 - \boldsymbol{v}_R.\hat{\boldsymbol{n}}\right) \tag{62}$$

where the minus sign comes from lowering the components of $\hat{\boldsymbol{n}}^i$ via $\eta_{ii}\, \hat{\boldsymbol{n}}^i$.

That Equation 62 gives us the first of an otherwise straightforward set of componentRules:

In[39]:=
```
ClearAll[componentRules];
componentRules = {
   r → R γR (1 - nDotV),
   λ[h_[0]] → γR,
   λ[h_[ud_]] :→ γR v[h[ud]],
   σ[h_[0]] → R,
   n[h_[0]] → 0,
   σ[h_[ud_]] :→ R n[h[ud]]};
```

Filter out acceleration terms, i.e., any terms depending on `dλds`,

In[41]:=
```
ClearAll[accelerationRules];
accelerationRules = {
   dλds[_] :→ 0};
```

In[43]:=
```
(FCoulomb = Fraw /. accelerationRules) /. dpyRules
```

Out[43]=

$$-\frac{e\,\lambda_\nu\left(\lambda_\mu - \frac{\sigma_\mu}{r}\right)}{r^2} + \frac{e\,\lambda_\mu\left(\lambda_\nu - \frac{\sigma_\nu}{r}\right)}{r^2}$$

$E^i = F^{i0}$ is the contravariant field vector we seek:

In[44]:=
```
(Fi0 = FCoulomb /. {μ → i, ν → 0}) /. dpyRules
```
Out[44]=

$$\frac{e\,\lambda_i\left(\lambda_0 - \frac{\sigma_0}{r}\right)}{r^2} - \frac{e\,\lambda_0\left(\lambda_i - \frac{\sigma_i}{r}\right)}{r^2}$$

But **Fraw** = $F_{\mu\nu}$ is doubly covariant. So, raising its two Latin (spacelike) indices $E^i = \eta^{i\mu}\,\eta^{0\nu}\,F_{\mu\nu} = -F_{i0}$:

In[45]:=
```
(Eraw = (-Fi0 /. componentRules) // Simplify) /. dpyRules
```
Out[45]=

$$\frac{e\,(n_i - v_i)}{(-1 + \text{nDotV})^3\,R^2\,\gamma_R^2}$$

This is equivalent to Carlip's Equation 1.8 after noting that $n_i - v_i = -(n^i - v^i)$. When there is no acceleration, $\hat{\boldsymbol{n}}_i - \boldsymbol{v}_R$ points along the line connecting the instantaneous position of the source and the test particle.

Conclusion and Future Work

This part of the notebook explores a $\mathcal{UD}$ (up-down) calculus for tensorial calculations, using the physical example of electrodynamical aberration as a proving ground. Our primary objective was to trade a bit of the brevity of standard component notation for the reliability of explicit valence tracking. We demonstrate that a "hybrid" approach—custom rules for contraction and gradients while relying on Wolfram's DisplayForm for readability—replicates a sophisticated result like Liénard-Wiechert field gradients without the ambiguity of manual index management. We verify that for a source moving with constant velocity, the velocity-dependent terms in the electromagnetic interaction exactly cancel the retardation effects, confirming that the electric field points toward the source's instantaneous position.

Future Work: Towards a $\mathcal{UD}$ DSL and Compiler

While the current prototype is adequate for our example prototype, several avenues for improvement remain if it is to approach a full-featured, embedded Domain Specific Language (eDSL) and compiler:

Automatic Dummy Index Management

Currently, we rely on the user to manually manage summation indices to avoid collisions, e.g., distinguishing the ρ in $r = \sigma_\rho\,\lambda^\rho$ from a free index μ. A implementation would automatically generate unique symbols for dummy indices during contraction. This is a standard feature in "alpha conversion."

Dummy Index Coalescing

The system should automatically recognize that $\sigma_\mu\,\lambda^\mu$ and $\sigma_\rho\,\lambda^\rho$ are identical scalars. For that matter, they're also identical to $\sigma^\mu\,\lambda_\mu$. Pattern matching that ignores the specific names of dummy indices and mixed parities in summations would afford better automatic simplification.

Native Operator Integration

We currently rely on a custom fourGrad operator. Future work could overload Wolfram's built-in Dt and D operators with UpValues that respect $\mathcal{UD}$ valence, allowing users to differentiate tensors as naturally as scalar variables without "stepping out" of the $\mathcal{UD}$ framework.

General Relativity and Curved Space

The rest of Carlip's paper extends electrodynamical aberration to gravitational aberration. To handle gravitation smoothly, we must extend fourGrad to the covariant derivative $\nabla\mu$, incorporating the Christoffel connection $\Gamma_{\mu\nu}{}^{\lambda}$. With the index tracking demonstrated here for electrodynamical aberration, deriving the Riemann curvature tensor and solving geodesic equations symbolically becomes a tractable next step.

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From the steam age to quantum fields: a unified approach via $\mathcal{UD}$ tensorial calculus
by Brian Beckman
Wolfram Community, STAFF PICKS, December 22, 2025
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